Sleep Scores: Your Smartwatch Is Not a Sleep Laboratory (And That's Fine)

The sleep score should answer “how restorative was last night’s sleep?”, not “am I ready to train?”. In practice this means sleep duration, continuity, regularity, and a small bit of physiology. The important point from the literature is that sleep trackers are usually more reliable for broad sleep-wake behaviour and trends than for exact metrics. Wrist actigraphy (method estimating sleep/wake using movement data from a wrist device) is strong for detecting sleep overall, but weaker for detecting wake, and consumer devices infer stages from movement and cardiorespiratory signals rather than measuring brain activity directly (obviously). [1]

We’ll start by understanding what the mainstream providers currently do to derive their sleep scores. They all follow a common pattern: mixing sleep amount, sleep quality/continuity, and some form of physiological recovery signal, but they differ in how much they expose the logic.

Fitbit’s sleep score has typically been described through dimensions such as duration, restoration, and restlessness, and Google’s more recent redesign appears to be moving toward a more transparent version that shows total sleep, time to reach “sound sleep,” interruptions, and full awakenings, in some cases with comparisons by age and gender. This is a sensible evolution because users can understand it (one of the bits of the health app I don't hate). [2]

Apple entered this space later with a Sleep Score in watchOS 26. Public reports say the score uses signals such as heart rate, wrist temperature, blood oxygen, and respiratory data, while other reporting indicates Apple also weights more actionable behavioural features like bedtime consistency and time in bed. The general direction is relatively clear: the company is treating sleep as both a physiological and a behavioural system. [3]

Garmin’s sleep ecosystem is less centered on one single number and more on the surrounding tools. Public guides emphasise overnight stress, Body Battery, respiration rate, and their Sleep Coach, which adjusts sleep goals based on activity, HRV, and naps. This serves as a useful reminder that a good sleep product is not just a number in the morning; it is also guidance on how much sleep the person likely needs, we want this number to help the user improve their readiness. [4]

WHOOP leans even harder into the “sleep as recovery fuel” framing. WHOOP directly links higher strain to higher Sleep Need, and the company’s Healthspan work treats sleep consistency and sleep duration as core long-term health metrics. [5]

Moving to the scientific literature, we see 3 large conclusions.

Firstly, sleep quantity and regularity are more robust than fine-grained sleep staging in wearables. Wrist actigraphy is good at identifying sleep versus wake overall, but it has difficulty with wake detection, and stage-level inference remains much less secure. More advanced wearable staging models using PPG and contextual windows can achieve good performance, but are still not the same as lab PSG (polysomnography - proper lab sleep studies). That doesn't make them useless but it means we should weight their outputs conservatively. [6]

The second is that sleep regularity matters far more than many apps used to admit. Large observational work summarised in recent reports found that irregular sleep schedules were strongly associated with worse cardiovascular outcomes, and other large-scale work linked higher sleep regularity to a lower mortality risk. From a product-design point of view, this means regularity should not be a tiny bonus metric but rather a major component of the score. [7]

The third is that wearables are most useful when they show trends and behaviour links, not when they pretend to be medical sleep labs. Sleep specialists quoted in recent reports make this point repeatedly: these devices can help people identify patterns around alcohol, late eating, bedtime drift, illness, and disturbance, but exact stage numbers should not be treated as truth. [8]

The more important question for us is not just "are wearables imperfect?", because they obviously are. The more useful question is which parts of sleep are reliable enough and important enough to drive a 'sleep score'. This is where I think the feature order becomes much clearer.

For newer products such as the Fitbit Air, we should be more optimistic than a generic "wrist trackers are bad" argument. Scientific evaluation videos compare the device against stronger reference measurements and suggest that these devices perform well enough to be useful, even if not perfect. The important distinction is that this should increase our confidence in wearables as a practical data source, but it should not change the basic scientific hierarchy. Even a good wearable is still more trustworthy for sleep duration, timing, continuity and broad sleep behaviour than it is for exact REM/deep/light stage breakdowns. [9]

The strongest order of importance looks like this:

- Sleep duration: This should likely be the largest part of the score. If the user does not get enough total sleep, the rest of the score should not be able to fully rescue it. Sleep stages can look "good" within a short sleep, but 4 hours of sleep is still, at the end of the day, 4 hours of sleep.
- Sleep regularity: This is the most underweighted feature in many current apps. Large cohort work has found sleep regularity to be strongly associated with mortality and cardiovascular outcomes, in some cases more strongly than just duration alone. From a scoring perspective, this means consistency should not be a tiny bonus. It should be a major block of the score. [10] [11]

- Continuity and awakenings: Sleep that is constantly interrupted is not the same as one continuous block of sleep. This is where wake after sleep onset, number of awakenings, restlessness, and sleep efficiency matter. Wearables are not perfect at wake detection, but this is still more reliable and actionable than obsessing over the exact stage percentages.
- Timing and sleep onset: Bedtime drift, very late sleep, and taking a long time to fall asleep should also matter, but slightly less than duration, regularity and continuity. Timing is important because it reflects circadian stability, not just "when did I happen to go to bed".
- Overnight physiology: HRV, RHR, respiratory rate, SpO_2 and temperature can help explain why sleep may have been poor or why the body looked stressed overnight. However, these should be smaller components of the sleep score than they are in a readiness score. They are more useful as context than as the definition of the sleep's quality.
- Sleep stages: REM and deep sleep should be included, but carefully. There is evidence that longer deep sleep, higher sleep efficiency and earlier REM latency are associated with more favourable wearable-derived health markers, but stage detection is still less secure than duration and timing. The key point is: sleep stages are useful supporting evidence, not the backbone of our score. [12] [13]

Given this, the score should weight the sleep score roughly like this: duration 35%, regularity 25%, continuity 20%, timing/onset 10%, overnight physiology 5%, and sleep stages 5%. If the wearable's stage tracking proves strong over time for a particular user, we can let stages have slightly more influence, but it would still be sensible to cap them around 10%. This keeps the score scientifically honest. It also makes the score more actionable because duration, regularity and continuity are things the user can actively improve.

The sleep score should be partly personalised but not in the same way as other scores. Duration should start from a general target, e.g. 7-9 hours for most adults, but then adapt slowly to the user's own sleep needs and recent strain. Regularity should be strongly personalised because the important thing is whether the user is consistent relative to their own life. Continuity should also be compared against their own baseline, because some people will naturally wake up more than others.

The score should also have a confidence layer. If duration and timing are available but stages are missing, the software should still calculate a sleep score as the strongest features are still present. If sleep stages look strange but duration, regularity and continuity are good, the score shouldn't collapse. If duration is short, bedtime is late, and awakenings are high, the score should drop even if the device claims a decent amount of REM. This is the core principle: the sleep score should reward the most reliable and most important sleep features first. This gives us a sleep score that answers the right question: "how restorative was last night's sleep?", rather than pretending the wearable is a medical sleep lab or accidentally turning the sleep score into a readiness score.

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